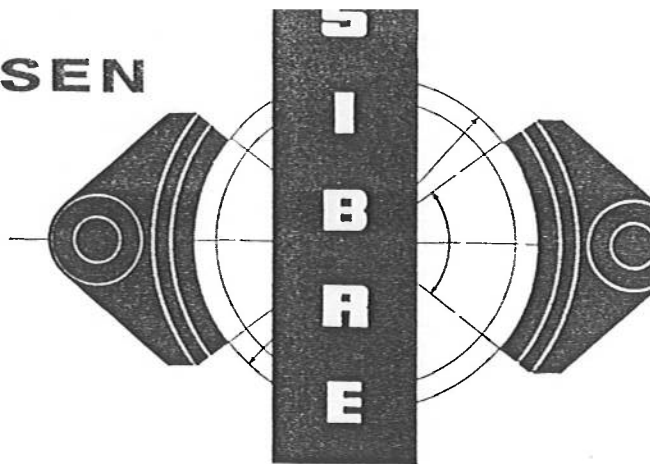


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HANDLING — MANUAL

of brake type SK 4 M

1. IN THE WORKSHOP

1.1. Solenoid operated Brake type SK 4 M

1.1.1. Check supply voltage on the nameplate
(see scheme)

1.1.2. Connect to a power supply and switch on for half an hour to heat up the coil. This is necessary as the magnetic thrust force decreases with rising temperature and the brake must be adjusted at its working temperature.

1.1.3. Loosen locking screw (66) on spring scale (65)

1.1.4. Adjusting of the spring tension adjusting screw (21)
-switch on and off to see if brake operates.

a) If the solenoid does not move, the spring tension is too high. Unscrew the adjusting screw (21) a quarter of a turn at a time until the solenoid can just overcome the resistance of the spring and travels the full 13 mm.

b) If the solenoid moves and operates the brake, now tighten the adjusting screw (21) a quarter of a turn at a time until the solenoid just overcomes the resistance of the spring and travels the full 13 mm.

Herewith you have set the brake for its max. braking capability.

1.1.5. Tighten up locking screw (66) on spring scale.

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Bahnstation:
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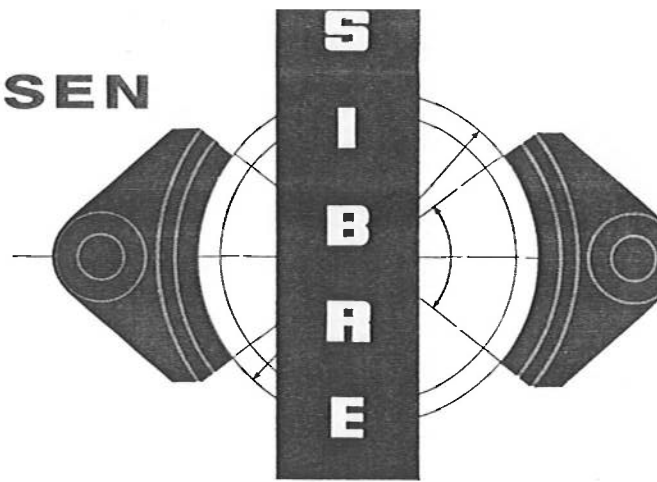
Geschäftsführer:
Ernst-Werner Kring
Dipl.-Ing. Roland Kring

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Persönlich haftende Gesellschafterin ist die Emde GmbH mit Sitz in 6342 Haiger-Rodenbach, Registergericht Dillenburg HRA 1300.

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2. INSTALLATION OF BRAKE ONTO DISC

- 2.1 Measure gap between shoes. It should be 1 to 2 mm greater than thickness of disc.
 - a) If gap is sufficient, see step 2.6
 - b) If gap is too small, take the following steps.
- 2.2 Slacken off screws (60)
- 2.3 Remove complete assembly (57). You can now see the adjusting nut (33) at one end of centreblock.
- 2.4 Insert a suitable steel rod into a hole in the edge of adjusting nut (33) and rotate the adjusting nut (33) toward brake disc until the gap between shoes is big enough and ensure the bore in sleeve (34) is uppermost. This is achieved by further rotation of the adjusting nut (33).
- 2.5 Carefully replace assembly (57) and tighten retaining screws (60)
- 2.6 Slide the brake onto the disc, place brake onto support bracket, insert (do not) tighten the 3 holding down setscrews.

Make sure these screws are not too long, otherwise the brake will be damaged and will not function!
- 2.7 Connect power supply (electric/pneumatic/hydraulic) incorporating normal overload protection and switching facilities.

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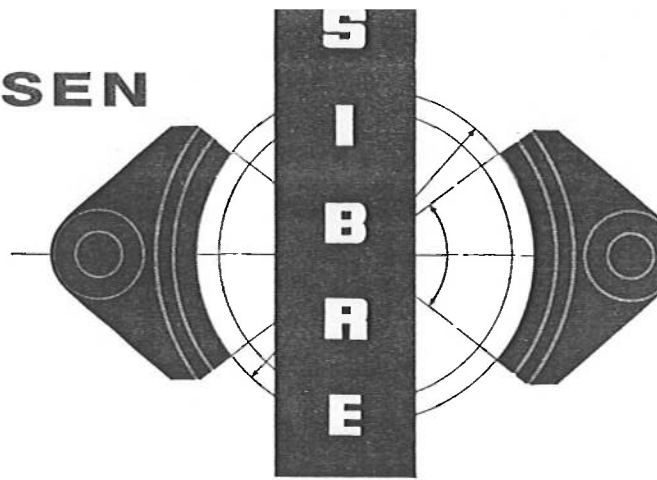
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- 2.8 Operate the brake several times to enable the automatic wear compensation device to put the centreblock in its correct position, which will be achieved when the safety gap between centreblock (30) and the guide tube (86/111/123) is as per following table:

Note: This safety gap can only be checked when the brake is in braking position.

	SK 4

Y	1,4 mm

- 2.9 This Safety Gap must be checked as follows:

After removal of the side plates (70) insert the correct thickness feeler gauge between the block and the guide tube.

For solenoid operated brakes it may be easier to check the free travel of the solenoid armature, which should be 5 mm.

- 2.10 Apply the brake and tighten the 3 holding down setscrews.

- 2.11 Screw in adjusting screws (4) until they touch the back of the shoe. Tighten up the locking nuts (5). This must be done when brake is applied.

Note: The brake lining and disc must be absolutely free of oil or grease!
The full braking torque will be achieved after some "wearing in" of the brake pads.

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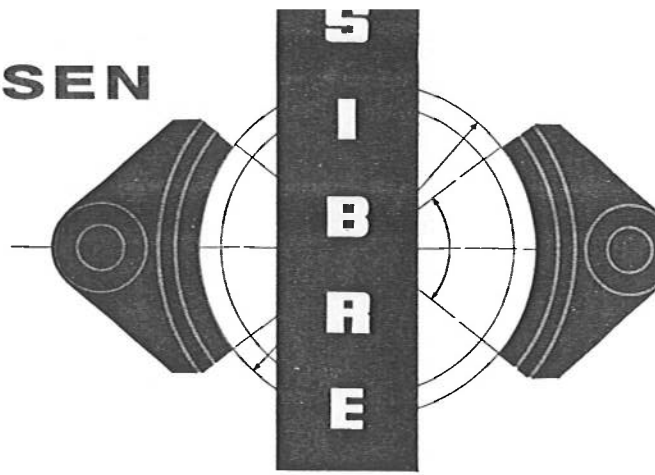
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3. REPLACEMENT OF BRAKE SHOES

- 3.1 Release the brake.
- 3.2 Slacken off screws (60)
- 3.3 Remove complete assembly (57). Now you can see the adjusting nut (33) at end of centreblock.
- 3.4 Insert a suitable steel rod into a hole in the edge of adjusting nut (33) and rotate the adjusting nut (33) toward brake disc until the gap between shoes is big enough and ensure the bore in sleeve (34) is uppermost. This is achieved by further rotation of adjusting nut (33).
- 3.5 Loosen and remove locknut (5) and setscrew (4)
- 3.6 Remove the two main locknuts (9) and washers (10)
- 3.7 Loosen locknut (7) and withdraw setscrew (6)
- 3.8 Remove main pivot pin (11)
- 3.9 Now remove the shoe
- 3.10 Remove old linings by removing setscrews (8) and replace new linings.

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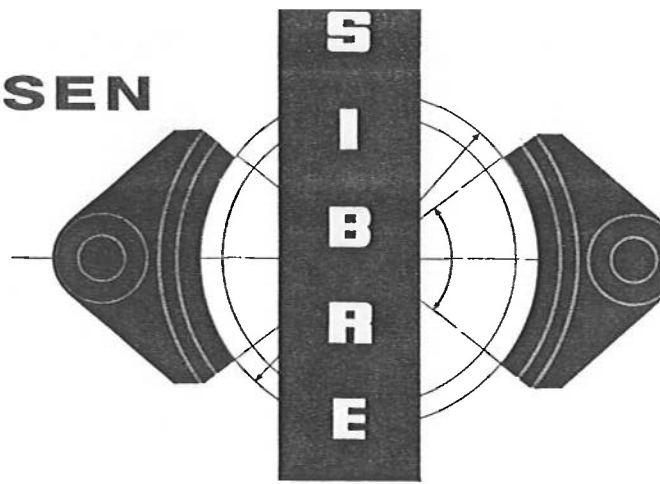
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- 3.11 Reassemble in reverse order
- make sure that setscrew (6) engages with dimple in pivot pin.
 - tighten up the main nuts (9) so that the cup springs (10) let the shoes just pivot.
- 3.12 While rotating the adjusting nut (33) backwards make sure that the gap between shoe and disc is 0.4 mm each side and also ensure the bore in sleeve (34) is uppermost.
- 3.13 Carefully replace assembly (57) and tighten retaining screws (60).
- 3.14 Operate the brake several times to enable the automatic wear compensation device to adjust the brake.

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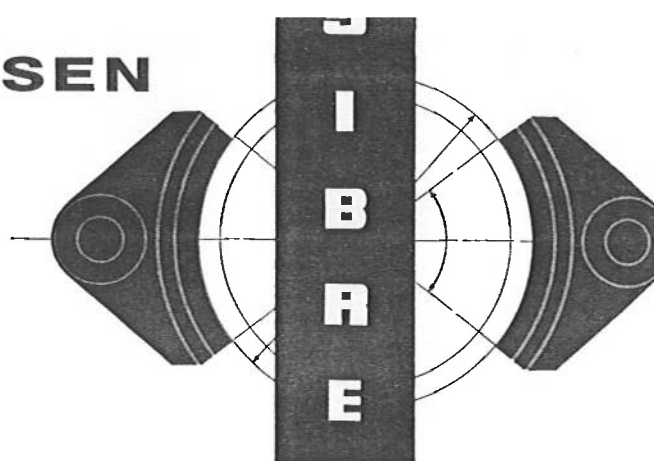
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4. ADJUSTMENT OF LEVER ON SOLENOID OPERATED BRAKES

- 4.1 Open up the gap between shoes, 1 mm each side by rotating the adjusting nut (33) as described in step 2.4
- 4.2 Switch the solenoid off.
- 4.3 Slacken locking nut (92) and unscrew setscrew (91)
- 4.4 Remove side covers
- 4.5 Centreblock (30) must be in contact with the guide tube (86)
- 4.6 Screw in the setscrew (91) until the lever is under tension, but make sure that the centreblock (30) still touches the guide tube (86).
- 4.7 Tighten locknut (92).
- 4.8 Operate brake several times to enable the automatic wear compensating device to adjust the brake.

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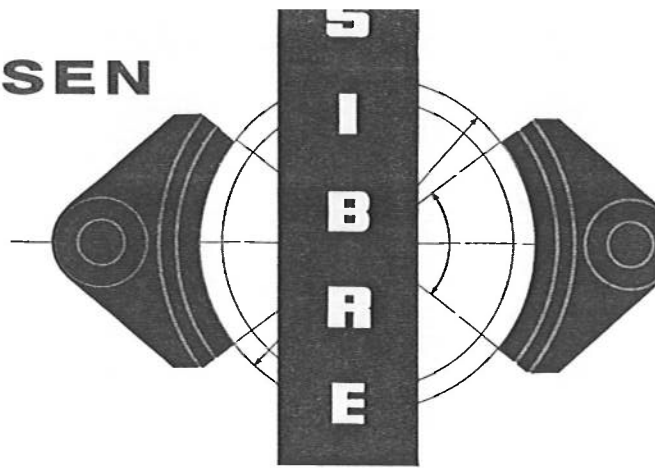
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5. ADJUSTMENT OF AUTOMATIC WEAR COMPENSATION DEVICE

- 5.1 For solenoid operated brakes, adjust the hand lever (as described in Parag. 4.)
- 5.2 Increase gap between shoes up to 1 mm (as described in steps 2.2 through 2.4)
- 5.3 Carefully replace assembly (57) and tighten retaining screws.
- 5.4 Slacken locknuts (53) and loosen setscrews (52)
- 5.5 Release brake by operating/energizing thrustor, thus moving the centreblock (30) forwards towards the brake disc.
The lever (54) will pivot towards the rear screw (52) which should be adjacent until both are just touching.
- 5.6 Tighten locknut (53).
- 5.7 Insert feeler gauge (Dimension "Y", Step 2.8) between guide tube (86) and centreblock (30) and release/de-energize thrustor.
- 5.8 The centreblock (30) is now in the braking position.

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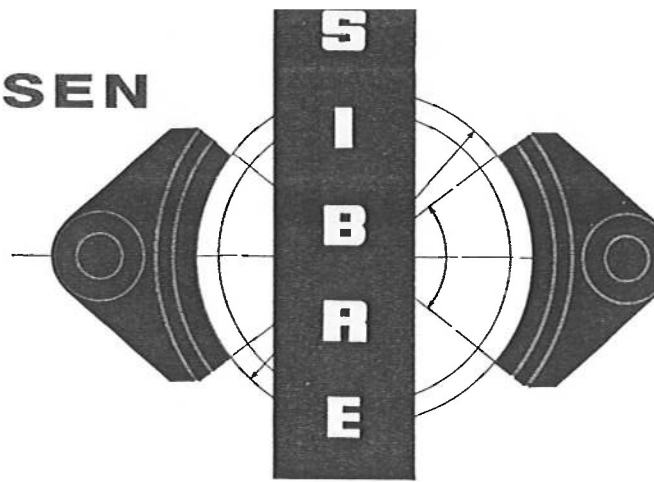
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- 5.10 Adjust setscrew (52) until it just touches lever (54) and tighten locknuts.
- 5.11 Energize the thruster and remove feeler gauge.
- 5.12 Actuate brake several times to enable the automatic wear compensation device to adjust brake. (For adjustments made in workshop use metal plate which is same width as disc).
- 5.13 Re-check dimension "Y"
- a) If dim. "Y" is too small, the setscrew (52) on the disc side should be tightend 1/4 turn after releasing locknut (53).
- b) If dim. "Y" is too large, the setscrew (52) on the disc side should be unscrewed 1/4 turn after releasing locknut (53).

In this case one must now repeat procedures as described in step 5.2

Actuate brake several times to enable the automatic wear compensating device to adjust brake and test functions.

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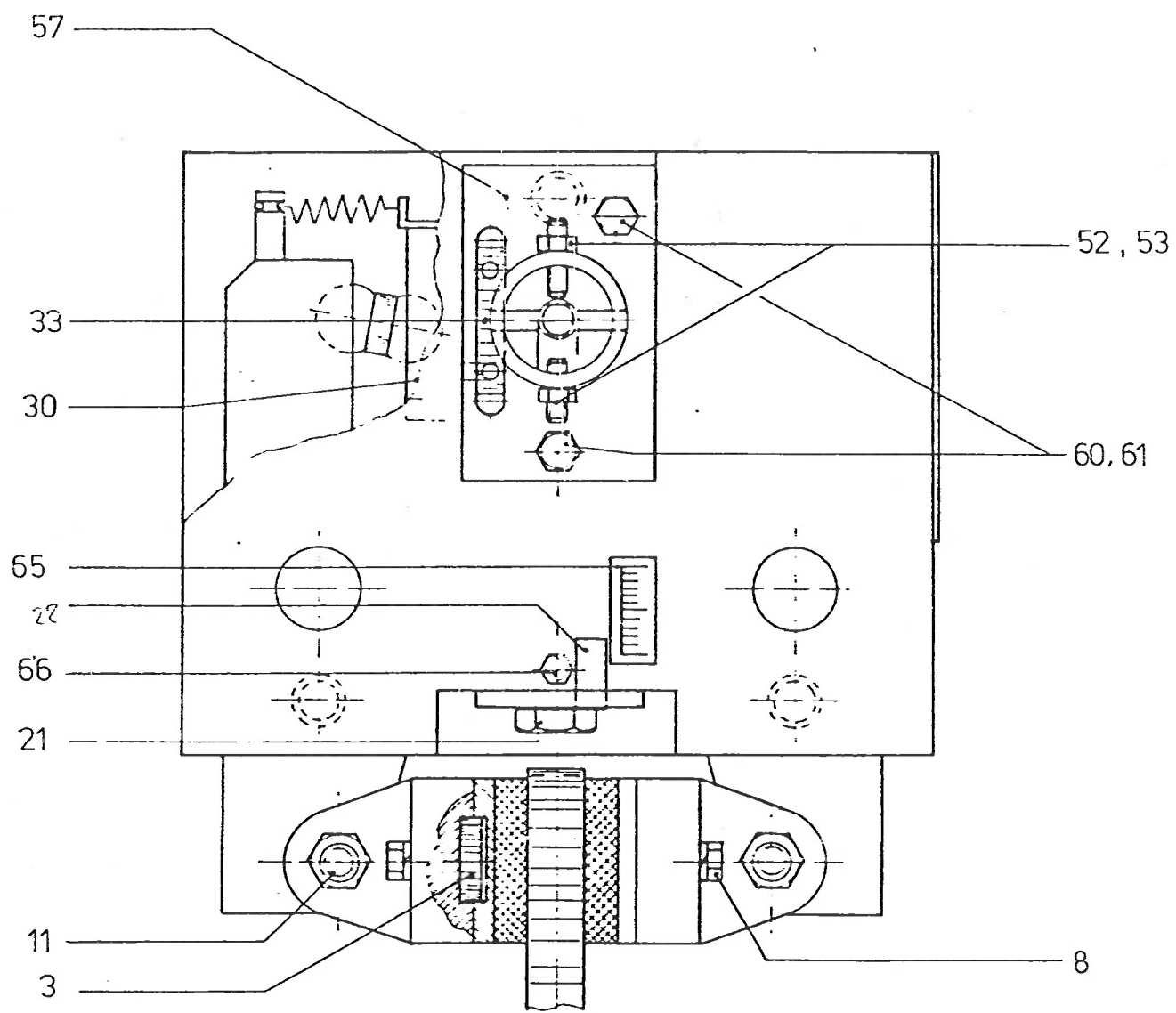


Figure 1

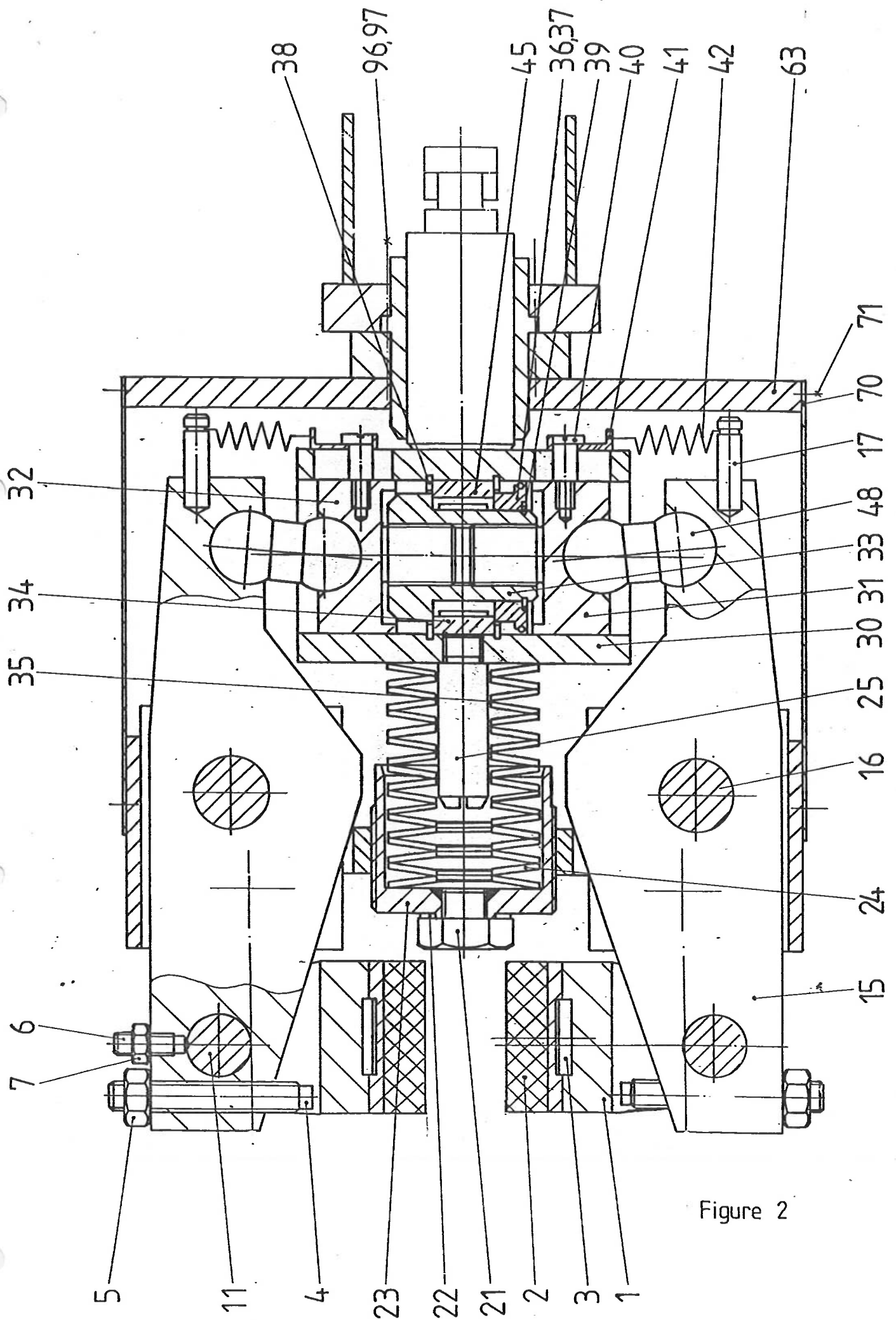


Figure 2

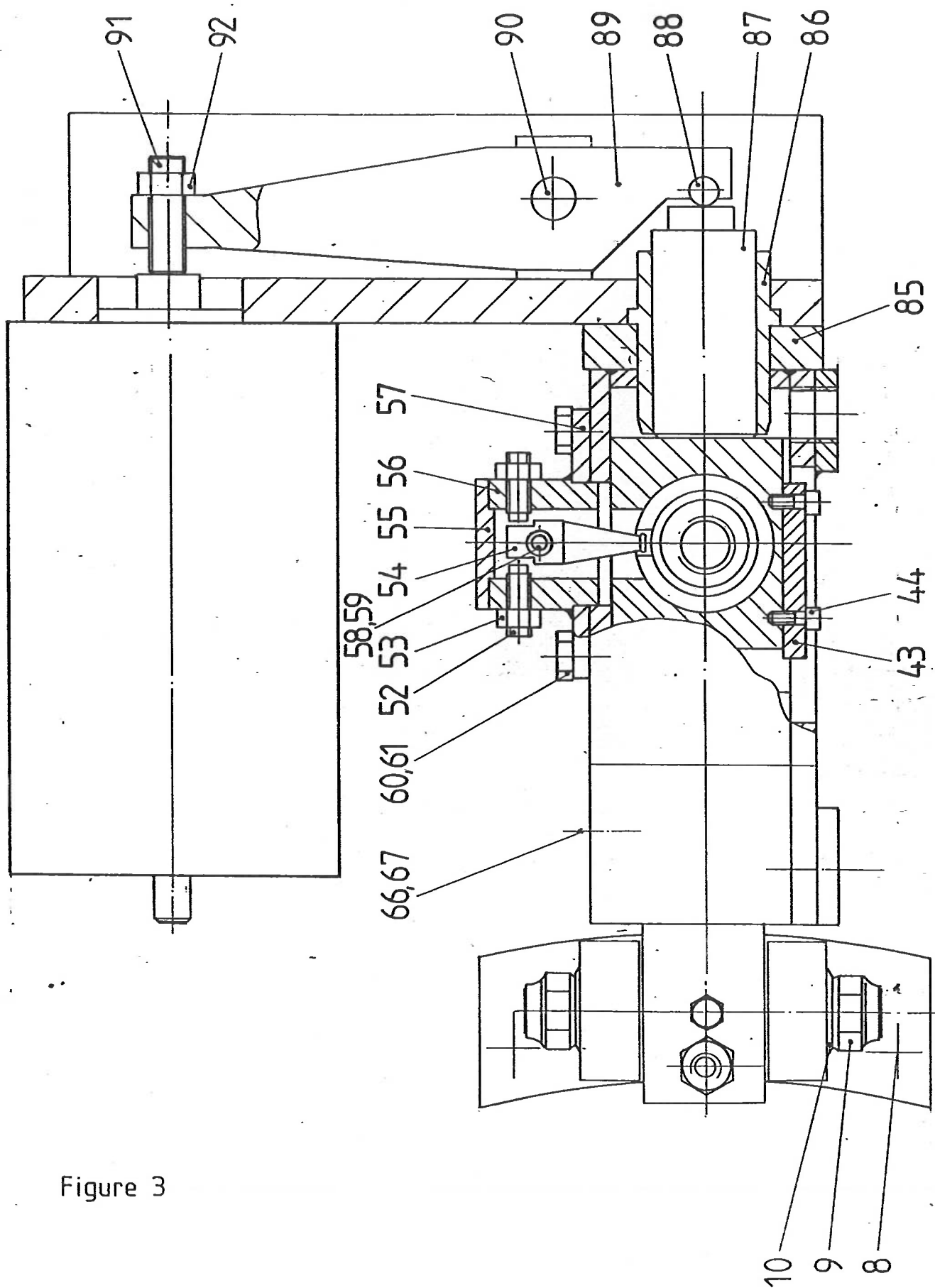
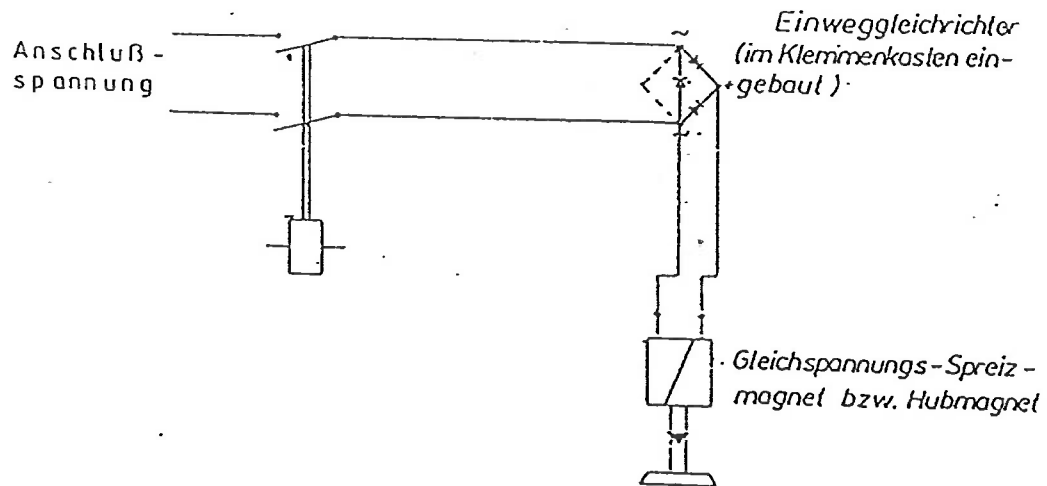


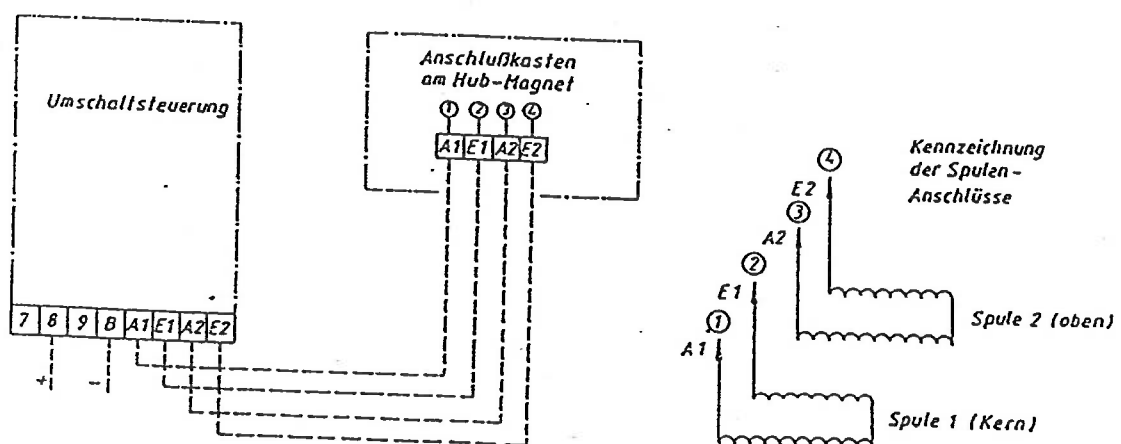
Figure 3



Schaltbild 1: (Hub- und Spreizmagnete)



Schaltbild 2: (nur Hubmagnete)



Schaltbild 3: (nur Hubmagnete)

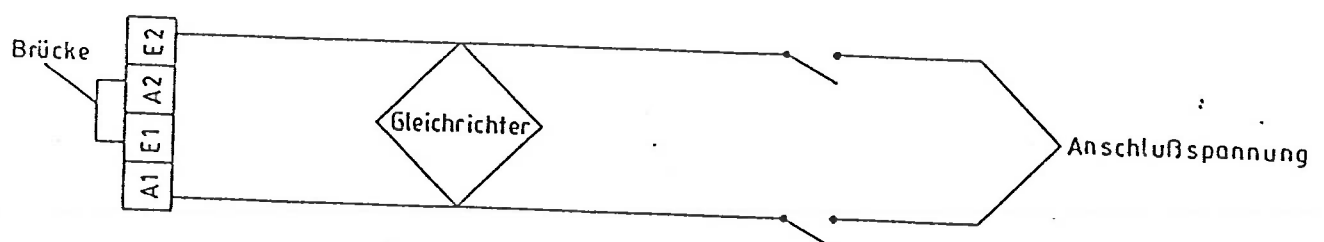
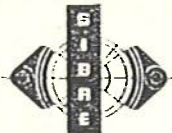


Abb.16: Anschlußpläne für Hub-(GH) und Spreizmagnete



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Werkszeugnis nach DIN 50049-2-2

Maker's Inspection Certificate according to DIN 50049-2-2

Relevé de Controle selon DIN 50049-2-2

Certificado de inspección del productor conforme a
DIN 50049-2-2

W 0 01 08 E

Besteller: Flender Singapore Pty Ltd
Orderer: 7 Jalan Rajawali
Commettant: Singapore, 2159
Comitente:

Bestell-Nr.:
Order No.:
Numéro de Commande:
Número de pedido: 457.211.001

Bremsentyp:
Brake Type:
Le type de frein:
Tipo de freno: 2 x SK4-M 13

Bestell-Datum:
Order Date:
Date de Commande:
Fecha del pedido: Febr. 1993

Bremsen-Nr.:
Brake No.:
Le numéro de frein:
Número del freno: 4454

Unsere Auftr.-Nr.:
Our Kom.-No.:
Notre Comm.:
Nuestro número del orden: 004454/D06001

Plan-Nr.:
Drawing No.:
Le numéro de plan:
Número de dibujo:

Bremsmoment:
Braking Torque:
Le couple de freinage: 295 Nm ($\mu = 0,35$)
Momento de freno:

Lieferbedingungen:
Delivery Conditions:
Conditions de livraison:
Condiciones de entrega: According to your order.

Art der Prüfung: Scheibenbremsen (disk brakes)

Bei den Trommelfremsen handelt es sich um ein Serienfabrikat, das während der Fertigung laufend kontrolliert wird.
Nach der Endmontage wurde eine Funktionsprüfung vorgenommen.

Kind of Testing:

These brakes are series products being tested continuously during the manufacture process.
A functional test is being made after the final assembly.

La manière de l'épreuve:

Avec le freins à tambour, il s'agit d'une réalisation en série qui est contrôlé couramment pendant la fabrication.
Après l'assemblage définitif un test de performance a été fait.

Modo de inspección:

Los frenos de tambor son artículos elaborados en serie que, durante la fabricación está continuamente inspeccionado.
Después del montaje final se hace una inspección funcional.

Wir bestätigen hiermit, daß die Lieferung den Vereinbarungen bei der Bestellung entspricht.

We herewith certify that the execution of your order has been effected in accordance with your contract.

Nous confirmons avec cela que la livraison correspond aux accords selon la commande.

Certificamos con esto que la entrega está conforme con las estipulaciones del pedido.

SIEGERLAND-BREMSEN
Emde GmbH & Co.

Haiger-Rodenbach, den 27.09.1993



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Werkzeugnis nach DIN 50049-2-2
Maker's Inspection Certificate according to DIN 50049-2-2
Relevé de Controle selon DIN 50049-2-2
Certificado de inspección del productor conforme a
DIN 50049-2-2

W 0 01 08 E

Besteller: Flender Singapore Pty Ltd
Orderer: 7 Jalan Rajawali
Committant: Singapore, 2159
Comitente:

Bestell-Nr.:
Order No.:
Número de Commande: 457.211.001
Número de pedido:

Bremsentyp:
Brake Type:
Le type de frein:
Tipo de freno: 6 x SK4-M-16

Bestell-Datum:
Order Date:
Date de Commande:
Fecha del pedido: Febr. 1993

Bremsen-Nr.:
Brake No.:
Le numéro de frein:
Número del freno: 4454

Unsere Auftr.-Nr.:
Our Kom.-No.:
Notre Comm.:
Nuestro número del orden: 004454/D06001

Plan-Nr.:
Drawing No.:
Le numéro de plan:
Número de dibujo:

Bremsmoment:
Braking Torque:
Le couple de freinage:
Momento de freno: 640 Nm ($\mu = 0,35$)

Lieferbedingungen:
Delivery Conditions:
Conditions de livraison:
Condiciones de entrega: According to your order.

Art der Prüfung: Scheibenbremsen (disk brakes)

Bei den ~~Discbremsen~~ handelt es sich um ein Serienfabrikat, das während der Fertigung laufend kontrolliert wird.
Nach der Endmontage wurde eine Funktionsprüfung vorgenommen.

Kind of Testing:

These brakes are series products being tested continuously during the manufacture process.
A functional test is being made after the final assembly.

La manière de l'épreuve:

Avec le freins à tambour, il s'agit d'une réalisation en série qui est contrôlé couramment pendant la fabrication.
Après l'assemblage définitif un test de performance a été fait.

Modo de inspección:

Los frenos de tambor son artículos elaborados en serie que, durante la fabricación está continuamente inspeccionado.
Después del montaje final se hace una inspección funcional.

Wir bestätigen hiermit, daß die Lieferung den Vereinbarungen bei der Bestellung entspricht.

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